Minerals vs Ores — Thermodynamic Stability, Terrestrial Abundance & Economic Viability

Elements occur in nature either in the **Native (Free) state** or in the **Combined state** depending on their electrochemical reactivity (E°). Highly unreactive noble metals (Au, Pt, and rarely Ag, Cu) occur native. Electropositive metals (s-block, Al, Zn, Fe) possess strongly negative oxidation potentials and occur exclusively as oxides, sulphides, carbonates, halides, or silicates.

1. Fundamental Definitions & The Thermodynamic Criterion

- **Mineral:** A naturally occurring, homogeneous inorganic substance having a definite chemical composition and crystalline structure found in the Earth's crust.
- **Ore:** A mineral from which one or more metals can be extracted **conveniently and profitably (economically viable)**. Hence: All ores are minerals, but all minerals are NOT ores!
- **Gangue (Matrix):** The non-metallic, undesirable earthy or rocky impurities (silica SiO_2, clay, silicates, limestone) contaminating the mined mineral ore.
- **Abundance Hierarchy in Earth's Crust:** By mass percentage: **O** (46.6%) > **Si** (27.7%) > **Al** (8.1%, most abundant metal) > **Fe** (5.0%, second most abundant metal) > **Ca** > **Na** > **K** > **Mg**.

2. Master Classification Table of Principal Ores (NCERT Table 6.1)

Metal	Principal Ore	Chemical Formula	Class / Type	Remarks & Traps
Aluminium	Bauxite Kaolinite (China clay)	$AlO_x(OH)_{3-2x}$ $(0 < x < 1)$ $Al_2Si_2O_5(OH)_4$	Oxide / Hydroxide Silicate clay	Corundum = α-Al_2O_3; Cryolite = Na_3AlF_6 Kaolinite is a mineral, not an economic ore
Iron	Haematite Magnetite Siderite Iron Pyrites	Fe_2O_3 Fe_3O_4 $FeCO_3$ FeS_2	Red oxide Magnetic mixed oxide Carbonate Disulphide	Iron pyrites ('Fool's Gold') is NOT used for Fe extraction due to high S content (used for H_2SO_4)
Copper	Copper Pyrites (Chalcopyrite) Malachite Cuprite Copper Glance (Chalcocite)	$CuFeS_2$ $CuCO_3{\cdot}Cu(OH)_2$ Cu_2O Cu_2S	Mixed sulphide Basic carbonate (Green) Red oxide (Ruby copper) Sulphide	Azurite = $2CuCO_3{\cdot}Cu(OH)_2$ (Blue) Chalcopyrite yields >70% of world's Cu
Zinc	Zinc Blende / Sphalerite Calamine Zincite	ZnS $ZnCO_3$ ZnO	Sulphide Carbonate Oxide	Sphalerite crystallizes in ccp ($Z=4$); Wurtzite is the hcp polymorph
Lead / Tin	Galena Cassiterite (Tinstone)	PbS SnO_2	Sulphide Oxide	Anglesite = $PbSO_4$; Cerussite = $PbCO_3$ Cassiterite contains magnetic wolframite impurity

Table 1.1 — Comprehensive ore reference matrix for JEE Advanced.

TRAP
CRITICAL TRAP: Iron Pyrites (FeS_2) is called 'Fool's Gold'. Even though it contains ≈46.6% iron, it is strictly NEVER used as an ore of iron in commercial metallurgy because the sulphur cannot be completely eliminated, leaving brittle iron (hot shortness). Instead, it is burned exclusively to produce SO_2 for sulphuric acid manufacture!

Concentration of Ores (Beneficiation / Dressing) — Physical Separation Techniques

Extraction begins by crushing the mined lumps in jaw crushers followed by pulverization in stamp/ball mills. Beneficiation removes gangue particles to enrich the metal concentration before chemical reduction. Physical methods exploit physical property disparities: density, magnetic susceptibility, or preferential wettability.

1. Hydraulic Washing (Gravity Separation / Levigation)

· **Physical Principle:** Based on the difference in **specific gravities (densities)** between the heavier ore particles and lighter gangue.
· **Apparatus & Working:** An upward stream of running water flows over a vibrating inclined table with wooden corrugations (Wilfley table) or hydraulic classifiers. The lighter gangue particles are washed away by the water stream, while the heavier ore grains settle in the grooves.
· **Applicability:** Heavily concentrated oxide and native ores: Haematite (Fe_2O_3, sp. gr. ≈5.2), Cassiterite (SnO_2, sp. gr. ≈7.0), and native Gold/Platinum.

2. Magnetic Separation

· **Physical Principle:** Exploits differences in **magnetic susceptibility** between the ore and the contaminating gangue particles (either the ore is magnetic and gangue non-magnetic, or vice-versa).
· **Apparatus:** Pulverized ore is dropped onto a moving conveyor belt stretched across two rollers, one of which houses a powerful electromagnet. Non-magnetic particles drop straight down under gravity into the outer receptacle; magnetic particles cling to the belt until it leaves the magnetic field, dropping into an inner bin.

Separation Case	Magnetic Component	Non-Magnetic Component	Metallurgical Context
Wolframite from Tinstone	Wolframite: $FeWO_4 + MnWO_4$ (Magnetic)	Cassiterite: SnO_2 (Diamagnetic)	Removal of heavy tungstate impurities from tin ore
Silica from Haematite	Haematite: Fe_2O_3 / Fe_3O_4 (Ferromagnetic)	Silica / Gangue: SiO_2 (Diamagnetic)	Upgrading low-grade iron ores
Rutile from Ilmenite	Ilmenite: $FeTiO_3$ (Magnetic)	Rutile: TiO_2 / Zircon (Non-magnetic)	Processing monazite beach sands of Kerala

Table 1.2 — Commercial magnetic separation cases tested in JEE Advanced.

KEY RESULT
Key Formula for Specific Gravity Differential: Separation efficiency $\eta \propto (\rho_{ore} - \rho_{fluid}) / (\rho_{gangue} - \rho_{fluid})$. If $\rho_{ore} \gg \rho_{gangue}$, gravity separation gives rapid, near-quantitative recovery.

Froth Floatation Process — Surface Chemistry of Preferential Wetting (NCERT §6.2.3)

Pioneered for concentrating **sulphide ores** (ZnS, PbS, $CuFeS_2$, Cu_2S). Physical basis: **Preferential wettability** — sulphide ore particles are preferentially wetted by non-polar oils (hydrophobic), whereas earthy gangue particles are wetted by water (hydrophilic).

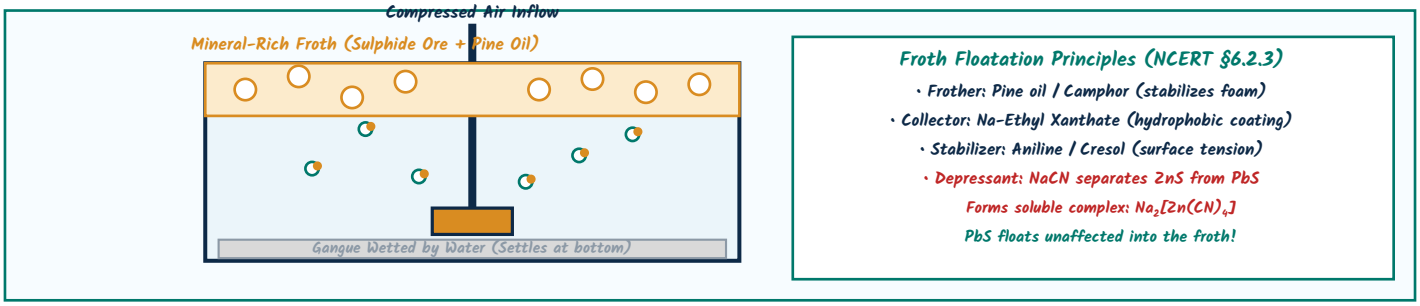


Fig. 1.1 — Froth floatation cell: agitator, rising mineral-laden bubbles, collectors, and depressant mechanism.

1. Reagents in Froth Floatation: Roles & Chemical Formulations

- **1. Frothing Agents (Frothers):** Lower the surface tension of water to produce a stable, persistent foam that carries minerals to the top. Examples: **Pine oil**, camphor oil, eucalyptus oil, and higher aliphatic alcohols.
- **2. Collectors (Hydrophobizing Agents):** Polar-nonpolar amphiphilic molecules whose polar heads chemisorb onto the metal cations of the sulphide surface, orienting their long non-polar hydrocarbon tails outward into the liquid, rendering the ore hydrophobic! Examples: **Sodium/Potassium ethyl xanthate** ($C_2H_5OCS_2^-Na^+$), potassium amyl xanthate, fatty acids.
- **3. Froth Stabilizers:** Molecules that stabilize the thin liquid lamellae of the bubbles, preventing premature collapse before skimming. Examples: **Aniline** ($C_6H_5NH_2$) and **Cresols** (o, m, p-$CH_3C_6H_4OH$).
- **4. Depressants (Selective Flotation):** Used when two sulphide ores co-exist in the same deposit (e.g. Galena PbS containing Zinc Blende ZnS). Adding **NaCN or KCN** selectively depresses ZnS by forming a water-soluble coordination complex: **$ZnS + 4\ CN^- \rightarrow [Zn(CN)_4]^{2-} + S^{2-}$**. The complexed zinc remains in solution, while unreacted PbS is collected in the froth! Later, adding an **activator** ($CuSO_4$) reactivates ZnS by precipitating insoluble CuS on its surface!

TRAP

EXAMINER TRAP: Depressant mechanism is a classic JEE question! In a mixture of PbS and ZnS, adding NaCN does NOT precipitate anything new; it dissolves ZnS as a stable cyanide complex $[Zn(CN)_4]^{2-}$. Galena (PbS) does NOT form a stable soluble cyanide complex under these conditions and floats freely.

Hydrometallurgical Leaching — Selective Chemical Dissolution (NCERT §6.2.4)

Leaching is used when the ore is soluble in a specific chemical reagent (acid, base, or complexing ligand) while the gangue impurities remain completely insoluble, or vice-versa. The dissolved metal is subsequently recovered by precipitation or displacement.

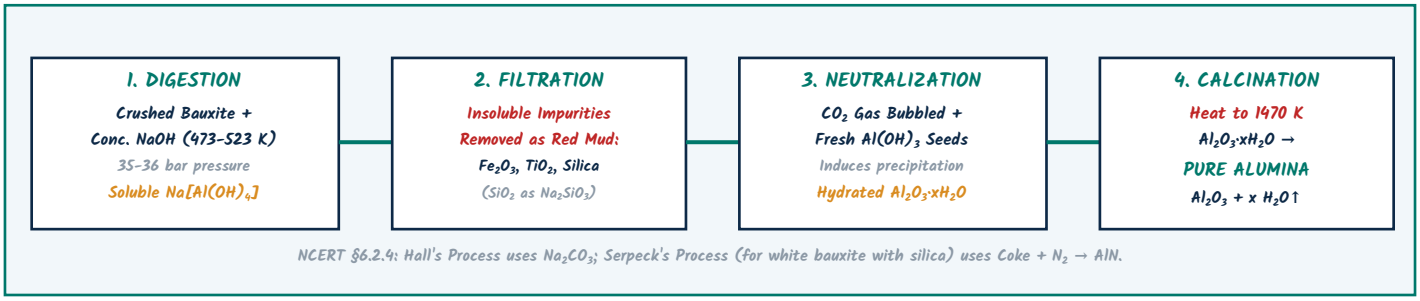


Fig. 1.2 — Bayer Process cycle: Digestion under pressure, red mud filtration, seeding neutralization, and calcination.

1. Leaching of Bauxite for Pure Alumina (Comparison of 3 Industrial Methods)

Process Name	Target Impurity in Bauxite	Leaching Reagent & Reaction	Precipitation & Calcination Steps
Bayer's Process (Red Bauxite)	Mainly Fe_2O_3 (red color) + minor SiO_2, TiO_2	Digested with conc. NaOH at 473–523 K, 35–36 bar: $Al_2O_3·2H_2O + 2NaOH + H_2O \rightarrow 2Na[Al(OH)_4]$ $SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$ (Red mud filtered)	Neutralized with CO_2 gas and seeded with fresh $Al(OH)_3$: $2Na[Al(OH)_4] + CO_2 \rightarrow 2Al(OH)_3 \downarrow + Na_2CO_3$ Calcinated at 1470 K: $Al(OH)_3$ / $Al_2O_3·xH_2O \xrightarrow{(\Delta)} Al_2O_3 + xH_2O$
Hall's Process (Red Bauxite)	Fe_2O_3, SiO_2	Fused with anhydrous Na_2CO_3 at ≈1273 K: $Al_2O_3 + Na_2CO_3 \rightarrow 2NaAlO_2 + CO_2 \uparrow$ Fe_2O_3 remains unreacted and filtered	Leach extract bubbled with CO_2 at 323–333 K: $2NaAlO_2 + CO_2 + 3H_2O \rightarrow 2Al(OH)_3 \downarrow + Na_2CO_3$ Followed by calcination to Al_2O_3 at 1470 K
Serpeck's Process (White Bauxite)	Mainly **SiO_2** (Silica-rich, iron-poor)	Heated with Coke + N_2 gas at ≈2073 K: $Al_2O_3 + 3C + N_2 \rightarrow 2AlN + 3CO \uparrow$ $SiO_2 + 2C \rightarrow Si \uparrow + 2CO \uparrow$ (Si volatilizes)	Hydrolysis of Aluminium Nitride: $AlN + 3H_2O \rightarrow Al(OH)_3 \downarrow + NH_3 \uparrow$ Valuable byproduct: Ammonia gas! Filtered & calcinated.

Table 1.3 — Complete comparative matrix of industrial bauxite leaching processes.

2. Mac-Arthur Forrest Cyanide Process for Gold & Silver

· **Dissolution (Oxidative Leaching):** Finely powdered native gold or argentite (Ag_2S) is treated with dilute (0.1–0.2%) NaCN in the presence of air (O_2 acts as the crucial oxidizing agent!):

$4\ Au + 8\ CN^- + 2\ H_2O + O_2 \rightarrow 4\ [Au(CN)_2]^- + 4\ OH^-$

$Ag_2S + 4\ CN^- \rightleftharpoons 2\ [Ag(CN)_2]^- + S^{2-}$ (O_2 oxidizes S^{2-} to SO_4^{2-} to drive equilibrium forward!)

· **Precipitation (Displacement):** The metal is recovered by adding electropositive **zinc dust**:

$2\ [Au(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2\ Au \downarrow$

Pyrometallurgical Pre-Treatment — Why Convert Ores to Oxides Prior to Reduction?

Sulphides, carbonates, and hydrated oxides are thermodynamically difficult to reduce directly to metals because their reduction Gibbs energies (ΔG°) are unfavorable, or they form volatile toxic gases. Metal oxides, however, possess well-characterized negative ΔG° curves that are readily reduced by carbon or CO at accessible temperatures!

1. Calcination vs Roasting — Definitive Diagnostic Comparison

Feature / Parameter	Calcination (Thermal Decomposition)	Roasting (Thermal Oxidation)
Atmosphere	Heated in **absence or limited supply of air** below its melting point	Heated in a **regular supply of excess air** below its melting point
Target Ores	Hydrated oxides, hydroxides, and **carbonates**	**Sulphide ores** exclusively (ZnS, PbS, $CuFeS_2$)
Chemical Reactions	Decomposition & dehydration: · $CaCO_3 \xrightarrow{(\Delta)} CaO + CO_2 \uparrow$ · $FeCO_3 \xrightarrow{(\Delta)} FeO + CO_2 \uparrow$ · $Al_2O_3{\cdot}2H_2O \xrightarrow{(\Delta)} Al_2O_3 + 2H_2O \uparrow$ · $CuCO_3{\cdot}Cu(OH)_2 \xrightarrow{(\Delta)} 2CuO + CO_2 + H_2O$	Oxidation of metal and non-metal impurities: · $2ZnS + 3O_2 \xrightarrow{(\Delta)} 2ZnO + 2SO_2 \uparrow$ · $2PbS + 3O_2 \xrightarrow{(\Delta)} 2PbO + 2SO_2 \uparrow$ · Impurity oxidation: $S + O_2 \rightarrow SO_2 \uparrow$; $P_4 + SO_2 \rightarrow P_4O_{10} \uparrow$; $4As + 3O_2 \rightarrow 2As_2O_3 \uparrow$ (volatile)
Physical Change	Ore becomes **porous and dry**, expulsion of crystal water and CO_2	Ore becomes **porous sintered oxide**; toxic SO_2 channeled to contact process
Furnace Employed	Reverberatory furnace or shaft kiln	Reverberatory furnace or multiple-hearth roaster

Table 1.4 — Calcination vs Roasting comparison.

KEY RESULT

Crucial Physical Benefit of Calcination/Roasting: The expulsion of gaseous products (CO_2, H_2O, SO_2) generates a vast network of internal micro-capillaries. This tremendous **increase in surface porosity** allows gaseous reducing agents (like CO) to rapidly penetrate deep into the ore matrix during blast furnace smelting!

TRAP

TRAP: Chloridizing Roasting vs Roasting — Roasting with NaCl converts refractory ores into water-soluble chlorides (e.g. $Ag_2S + 2NaCl \rightarrow 2AgCl + Na_2S$), used prior to cyanidation. Do not confuse standard oxidative roasting with chloridizing roasting!

Flux-Gangue-Slag High-Temperature Equilibrium — High-Temperature Lewis Acid-Base Reactions

Even after physical beneficiation, some non-fusible gangue impurities remain embedded in the ore. A **Flux** is a chemical substance added during smelting to react with non-fusible gangue at elevated temperatures, converting it into a **fusible (low-melting), light molten liquid called SLAG**.

1. The Slag Equation: Gangue + Flux = Slag

· **Physical Properties of Slag:**
1. **Low Melting Point:** Fusible at furnace operating temperatures.
2. **Low Density:** Significantly lighter than the molten metal; forms an immiscible floating blanket over the molten metal bath.
3. **Protective Blanket:** The floating slag layer shields the hot molten metal from aerial re-oxidation by atmospheric O_2!

2. Classification of Fluxes & Slag Chemistry

Gangue Nature	Nature of Flux Required	Flux Substance Added	Slag Formation Reaction	Metallurgical System
Acidic Gangue (Excess SiO_2, P_4O_{10})	**Basic Flux**	Limestone $CaCO_3$ (decomposes to CaO), $MgCO_3$, or $FeCO_3$	$CaO + SiO_2 \rightarrow CaSiO_3$ (Calcium silicate slag) $6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$ (Thomas slag)	**Blast Furnace extraction of Iron** (Haematite has quartz SiO_2 gangue)
Basic Gangue (Excess FeO, CaO, MgO)	**Acidic Flux**	Silica sand (SiO_2) or Borax ($Na_2B_4O_7$)	$FeO + SiO_2 \rightarrow FeSiO_3$ (Ferrous silicate slag) $CaO + SiO_2 \rightarrow CaSiO_3$	**Extraction of Copper** from chalcopyrite (roasting produces FeO gangue!)
Neutral Flux	**Neutral / Fluidity Aid**	Fluorspar (CaF_2), Cryolite (Na_3AlF_6)	Does not form new chemical compound; lowers melt viscosity and melting point	**Hall-Héroult Process** for Alumina

Table 1.5 — Flux and slag systems in extractive metallurgy.

KEY RESULT
Thomas Slag (Agricultural Fertilizer): The slag produced during steel manufacturing in the basic Bessemer process contains $Ca_3(PO_4)_2 \cdot CaSiO_3$. Known as **Thomas Slag**, it is pulverized and sold directly as a rich phosphorus fertilizer for agriculture!

Thermodynamic Foundations of Pyrometallurgy — Ellingham Diagrams (NCERT §6.3)

For any metallurgical reduction to proceed spontaneously at temperature T, the overall change in standard Gibbs free energy must be negative: $\Delta G^\circ = \Delta H^\circ - T\,\Delta S^\circ < 0$. An **Ellingham diagram** is a graphical plot of $\Delta_r G^\circ$ versus Absolute Temperature T for the formation of metal oxides per mole of consumed oxygen: $(2x/y)\ M + O_2\ (g) \rightarrow (2/y)\ M_xO_y(s)$.

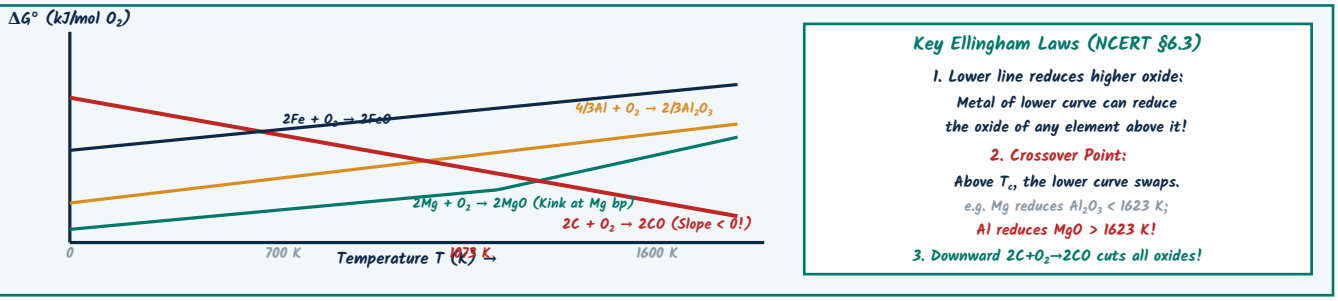


Fig. 2.1 — The Ellingham Diagram: Slopes, phase change kinks, crossover temperatures, and C/CO lines.

1. Crucial Features of the Ellingham Diagram

- **1. Positive Slope for Metal Oxides:** Since $O_2(g)$ (a highly disordered gas) is consumed to form a dense, ordered solid oxide, entropy decreases: $\Delta S^\circ < 0$. Because slope = $d(\Delta G^\circ)/dT = -\Delta S^\circ$, the slope is **positive (+ve)**! Lines slope upward from left to right.
- **2. Abrupt Upward Kink upon Phase Transition:** When the metal or oxide melts or vaporizes, ΔS° becomes even more negative (gas to liquid/solid involves huge entropy drop). Hence, the slope abruptly steepens (+ve slope increases) at the melting/boiling point of the metal (e.g. sharp upward kinks at the boiling points of Mg at 1390 K and Zn at 1180 K)!
- **3. Feasibility Criterion:** Any metal whose oxide line lies **LOWER** on the Ellingham diagram can reduce the oxide of any metal lying **HIGHER** on the diagram at that temperature! (Because reversing the higher reaction and adding yields a net negative $\Delta_r G^\circ$).

TRAP
CRITICAL TRAP: Ellingham diagrams provide purely **THERMODYNAMIC FEASIBILITY**; they give ZERO information regarding the **KINETICS (rate of reaction)**! A reaction may have a massive negative ΔG° at room temperature but possess an immense activation energy (E_a), making it imperceptibly slow unless heated to operating temperature!

The Carbon-Oxygen System in Ellingham Space — Why Carbon Reduces Almost Everything

In the Ellingham diagram, carbon exhibits three distinct oxidation lines with wildly different slopes, governed strictly by the stoichiometry of gas moles formed versus consumed.

1. Analysis of the Three Carbon Curves

Oxidation Reaction	Stoichiometry & Entropy Change	Slope on Ellingham Plot	Thermodynamic Consequence
$C(s) + O_2(g) \rightarrow CO_2(g)$	1 mole gas in → 1 mole gas out: $\Delta S^\circ \approx 0$	**Nearly horizontal** (zero slope)	Line remains flat across all temperatures (≈−394 kJ/mol)
$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$	3 moles gas in → 2 moles gas out: $\Delta S^\circ < 0$	**Positive slope** (slopes upward)	Strong reducing agent only at **low temperatures** ($T < 1073$ K)
$2C(s) + O_2(g) \rightarrow 2CO(g)$	1 mole gas in → 2 moles gas out: $\Delta S^\circ > 0$!	**NEGATIVE slope** (slopes downward!)	As T increases, ΔG° becomes **increasingly negative**!

Table 2.1 — The three carbon oxidation curves in Ellingham diagrams.

2. The Carbon Crossover Point at 1073 K (Boudouard Equilibrium)

At approximately **1073 K (800°C)**, the curves for $C \rightarrow CO$ and $C \rightarrow CO_2$ intersect. This temperature represents the fundamental crossover boundary:

· **Below 1073 K:** The oxidation of CO to CO_2 lies lower than $C \rightarrow CO_2$. Therefore, **CO is a thermodynamically stronger and kinetically superior reducing agent than carbon**! This explains why in the upper (cooler) zone of the blast furnace, iron oxides are reduced by CO, not C!

· **Above 1073 K:** The curve for $2C + O_2 \rightarrow 2CO$ dives continuously downwards. Therefore, **Carbon (coke) becomes a dramatically stronger reducing agent than CO**, and directly reduces FeO in the high-temperature hearth!

KEY RESULT

Boudouard Equilibrium: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ ($\Delta H > 0$). Le Chatelier's principle dictates that high temperature favours endothermic CO formation, while low temperature favours CO_2.

Carbon Smelting — Industrial High-Temperature Reduction of Intermediate Oxides

Carbon reduction (smelting) is applied to moderately electropositive metals whose oxides lie above the downward-sloping carbon line at attainable furnace temperatures (1000 – 1800 K).

1. Carbon Reduction of Zinc Oxide (Extraction of Zinc)

- **Ore & Roasting:** Sphalerite (ZnS) is roasted to Zincite: $2ZnS + 3O_2 \xrightarrow{(1200\ K)} 2ZnO + 2SO_2 \uparrow$.
- **Reduction Reaction:** Crushed ZnO is mixed with powdered coke and heated to ≈1673 K in vertical retort furnaces: $\mathbf{ZnO(s) + C(s) \xrightarrow{(1673\ K)} Zn(g) + CO(g)}$.
- **Crucial Physical Detail:** Zinc boils at **1180 K (907°C)**! At the reduction temperature (1673 K), zinc is formed in the **vapour state**! The vapours are rapidly condensed by cooling to prevent re-oxidation by CO.
- **Spelter:** The condensed crude zinc is ≈97–98% pure, known commercially as **Spelter** (contains Pb, Fe, Cd impurities).

2. Carbon Reduction of Cassiterite (Extraction of Tin)

· Roasted tinstone (SnO_2) is mixed with anthracite coal and limestone flux and smelted in a reverberatory furnace at 1500 K:

$\mathbf{SnO_2 + 2C \xrightarrow{(1500\ K)} Sn(l) + 2CO} \uparrow$

· Molten tin is tapped off, leaving slag containing residual tin ('tin slag'), which is re-smelted with more lime and coke.

Metal Oxide	Operating Temp	Physical State of Metal	Furnace Type	Refining Requirement
ZnO	≈1673 K	**Vapour** (b.p. 1180 K)	Vertical retort / Belgo-furnace	Fractional distillation (removes Pb, Cd)
SnO_2	≈1500 K	**Molten liquid**	Reverberatory furnace	Liquation followed by poling
PbO	≈1200 K	**Molten liquid**	Blast / Reverberatory furnace	Parke's / Pattinson's process (desilverization)

Table 2.2 — Industrial carbon reduction parameters for non-ferrous metals.

Auto-Reduction (Self-Reduction) — Reduction Without External Reducing Agents (NCERT §6.4)

Applied exclusively to the sulphide ores of metals having **relatively low electropositivity** and intermediate affinity for oxygen and sulphur: **Copper (Cu), Lead (Pb), and Mercury (Hg)**. No carbon or external flux is added for reduction!

1. The Fundamental Self-Reduction Principle

The concentrated sulphide ore is subjected to **partial roasting** in air to convert a portion of it into the corresponding metal oxide or sulphate. Air supply is then cut off, and the temperature is raised. The newly formed oxide/sulphate reacts with the remaining unoxidized sulphide ore to yield the free metal and SO_2 gas!

2. Industrial Self-Reduction Case Studies

Metal	Ore & Formula	Step 1: Partial Roasting (with Air)	Step 2: Self-Reduction (Air Cut Off)
Copper (Cu)	Chalcocite / Copper Glance (Cu_2S)	$2Cu_2S + 3O_2 \xrightarrow{(\Delta)} 2Cu_2O + 2SO_2 \uparrow$	**$2Cu_2O + Cu_2S \xrightarrow{(\Delta)} 6Cu(l) + SO_2 \uparrow$** Occurs in Bessemer converter; produces Blister Copper!
Lead (Pb)	Galena (PbS)	$2PbS + 3O_2 \xrightarrow{(\Delta)} 2PbO + 2SO_2 \uparrow$ $PbS + 2O_2 \xrightarrow{(\Delta)} PbSO_4$	**$2PbO + PbS \xrightarrow{(\Delta)} 3Pb(l) + SO_2 \uparrow$** **$PbSO_4 + PbS \xrightarrow{(\Delta)} 2Pb(l) + 2SO_2 \uparrow$** Occurs on the hearth of a reverberatory furnace.
Mercury (Hg)	Cinnabar (HgS)	$2HgS + 3O_2 \xrightarrow{(\Delta)} 2HgO + 2SO_2 \uparrow$	**$2HgO + HgS \xrightarrow{(\Delta)} 3Hg(g) + SO_2 \uparrow$** (Also direct thermal decomposition: $HgO \xrightarrow{(\Delta)} Hg + \frac{1}{2}O_2$; Hg boils at 630 K and is distilled).

Table 2.3 — Self-reduction stoichiometry for Cu, Pb, and Hg in JEE Advanced.

TRAP

TRAP: Can Zinc or Iron undergo self-reduction? ABSOLUTELY NOT! The oxides ZnO and FeO are far too thermodynamically stable compared to ZnS and FeS (their $\Delta_f G^\circ$ lines lie deep below the sulphide lines). Thus, $ZnO + ZnS \rightarrow 2Zn + SO_2$ has a heavily positive ΔG° and cannot occur under any metallurgical conditions!

Thermite Process & Displacement Reduction — Highly Exothermic Metallurgical Reactions

When an oxide is too stable to be reduced by carbon, or when the metal reacts with carbon to form brittle interstitial carbides (like Cr_3C_2, Mn_3C, TiC), more electropositive metals (Al, Mg, Na) are deployed as reducing agents.

1. Goldschmidt Thermite Process (Aluminothermic Reduction)

· **Physical Basis:** Aluminium has an extraordinarily high affinity for oxygen due to the massive negative standard enthalpy of formation of alumina: Δ_f **$H°(Al_2O_3) = -1676$ kJ/mol**!
· **Charge & Ignition:** The metal oxide is mixed with granular aluminium powder (3:1 ratio called **Thermite**), placed in a fire-clay crucible embedded in sand, and ignited using an ignition mixture of **Barium peroxide (BaO_2) + Mg ribbon**:
$Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr(l)$ ($\Delta H = -540$ kJ/mol, T > 3000 K!)
$3Mn_3O_4 + 8Al \rightarrow 4Al_2O_3 + 9Mn(l)$
$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe(l)$ (Thermite welding of railway tracks!)
· The reaction generates so much heat that the reduced metal is obtained in the **molten liquid state**, while lighter Al_2O_3 floats cleanly on top as slag.

2. Metal Displacement & Hydrometallurgy of Low-Grade Ores

· Low-grade copper ores and tailings (CuO, $CuCO_3$) are leached with dilute H_2SO_4 to yield aqueous $CuSO_4$.
· The solution is treated with scrap iron or hydrogen gas to displace copper metal economically:
$Cu^{2+}(aq) + Fe(s) \rightarrow Cu(s) + Fe^{2+}(aq)$ ($\Delta E° = +0.78$ V)
$Cu^{2+}(aq) + H_2(g) \rightarrow Cu(s) + 2H^+(aq)$

KEY RESULT
Why is scrap iron preferred over scrap zinc to precipitate copper? Even though Zinc is more electropositive ($E° = -0.76$ V), Zinc is vastly more expensive than scrap iron! Scrap iron provides cheap, high-yield displacement.

Electrolytic Reduction of Halides & Oxides — Taming s-Block & Reactive Metals (NCERT §6.4)

Metals with strongly negative reduction potentials (Na, K, Mg, Ca, Al) CANNOT be reduced by chemical agents (carbon forms carbides, hydrogen fails, Ellingham lines are at the very bottom). Furthermore, they **CANNOT be extracted by electrolyzing their aqueous solutions** because H_2O is reduced preferentially to H_2 gas ($E^\circ = -0.83$ V) instead of the metal! Hence, **FUSED (MOLTEN) SALT ELECTROLYSIS** is mandatory.

1. Industrial Electrometallurgical Processes

Metal	Industrial Process	Fused Electrolyte Composition	Electrode Reactions & JEE Traps
Sodium (Na)	**Down's Process**	Molten NaCl (40%) + $CaCl_2$ (60%)	**Why add $CaCl_2$?** Pure NaCl melts at 1074 K (where Na vaporizes and forms a fog with melt). Adding $CaCl_2$ lowers m.p. to ≈873 K! · Cathode (Iron): $Na^+ + e^- \rightarrow Na(l)$ · Anode (Graphite): $2Cl^- \rightarrow Cl_2(g) + 2e^-$
Magnesium (Mg)	**Dow's Sea Water Process**	Anhydrous $MgCl_2$ (35%) + NaCl (50%) + $CaCl_2$ (15%)	Carnallite ($KCl{\cdot}MgCl_2{\cdot}6H_2O$) cannot be dehydrated by direct heating (hydrolyzes to MgO + 2HCl). It must be dehydrated in a stream of dry HCl gas! · Cathode (Steel): $Mg^{2+} + 2e^- \rightarrow Mg(l)$
Aluminium (Al)	**Hall-Héroult Process**	Molten Al_2O_3 (5%) + Na_3AlF_6 (85%) + CaF_2 (10%)	Pure Al_2O_3 melts at 2320 K and is a poor conductor. Cryolite lowers m.p. to 1170 K and provides ionic conductivity. · Cathode (Carbon lining): $Al^{3+} + 3e^- \rightarrow Al(l)$ · Anode (Graphite): $2O^{2-} + C \rightarrow CO_2 + 4e^-$

Table 2.4 — Master Electrometallurgical processes and electrolyte additives.

TRAP
EXAMINER TRAP: In Down's process, Ca^{2+} has a more negative reduction potential than Na^+ under operating conditions, so ONLY sodium is deposited at the cathode; trace Ca that codeposits forms an immiscible crust that is easily separated.

Purification of Crude Metals — Thermal, Physical & Chemical Refining (NCERT §6.5)

Metals obtained after reduction are 'crude' and contain unreacted ore, other metals, non-metals (C, Si, P, S), or slag. Refining upgrades purity to >99.9%. The choice of technique depends on differences in physical or chemical properties between the metal and its impurities.

1. Distillation Refining

- **Principle:** Applied to metals with **low boiling points** contaminated with non-volatile impurities.
- **Technique:** Crude metal is heated above its boiling point; the pure metal vapours distill off and are condensed in a cooled receiver.
- **Applicable Metals: Zinc** (b.p. 1180 K), **Cadmium** (b.p. 1040 K), and **Mercury** (b.p. 630 K).

2. Liquation Refining

- **Principle:** Applied when the **metal has a much lower melting point** than the contaminating impurities.
- **Technique:** Crude metal is placed on the sloping hearth of a reverberatory furnace heated slightly above the metal's melting point. The low-melting pure metal melts and flows down the slope into a collection well, leaving the solid, infusible dross behind.
- **Applicable Metals: Tin (Sn)** (m.p. 505 K, separated from high-melting iron dross) and **Lead (Pb)** (separated from copper and silver dross).

3. Poling (Green Wood Refining)

- **Principle & Technique:** Used when the crude metal contains **its own oxide as the primary impurity** (e.g. Cu_2O in blister copper, or SnO_2 in tin). The molten crude metal is vigorously stirred with **poles of fresh green wood**.
- **Mechanism:** The intense heat causes the green wood to release reducing hydrocarbon gases (chiefly **Methane CH_4**), which chemically reduce the metal oxide back to free metal without introducing new impurities:

$3\ Cu_2O + CH_4 \rightarrow 6\ Cu + CO\uparrow + 2\ H_2O\uparrow$

$SnO_2 + CH_4 \rightarrow Sn + CO\uparrow + 2\ H_2O\uparrow$

4. Cupellation Refining

- Used to refine crude Silver or Gold containing **Lead (Pb)** impurity (from lead-silver ores).
- Molten crude alloy is heated in an oval porous cupel made of bone-ash or cement in a blast of air. Lead oxidizes rapidly to volatile litharge (PbO), which vaporizes and is absorbed by the porous bone-ash walls, leaving a gleaming bead of pure silver!

TRAP
TRAP: Poling reduces oxides back to metal; it does NOT remove non-metal impurities like sulphur or phosphorus!

Ultra-High Purity Refining — Semiconductor Grade Materials & Volatile Complexes

Certain high-technology applications (semiconductors, solar cells, aerospace alloys) demand extreme purity levels exceeding **99.9999% (six-nines purity)**. Conventional thermal methods cannot reach this threshold.

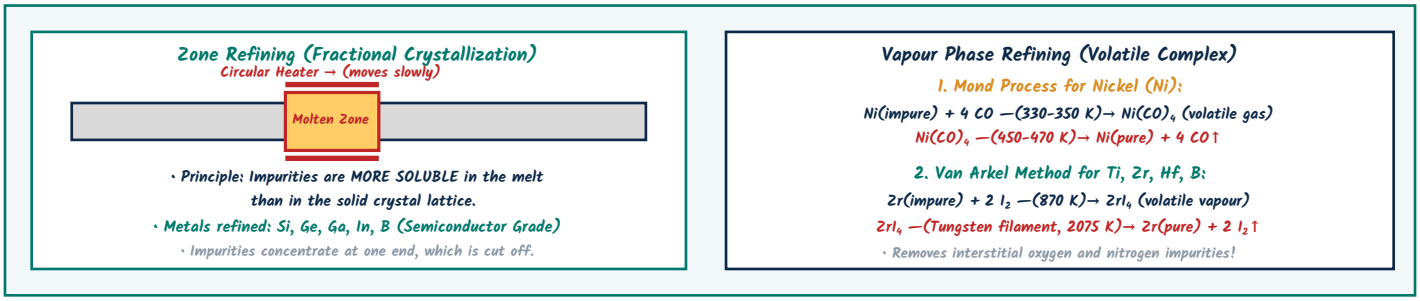


Fig. 3.1 — High-Purity Refining: Zone refining mechanism and Vapour Phase complexes (Mond and Van Arkel).

1. Zone Refining (Fractional Crystallization Principle)

- **Underlying Principle:** Based on the fundamental thermodynamic fact that **impurities are significantly MORE SOLUBLE in the molten liquid state than in the corresponding solid crystal lattice** ($K_D = C_S / C_L < 1$).
- **Working Mechanism:** A circular mobile induction heater surrounds a rod of impure metal enclosed in an inert atmosphere (argon) and travels slowly from one end to the other. As the molten zone sweeps forward, pure metal recrystallizes behind it, while impurities stay segregated in the molten zone!
- **Repetition & Truncation:** The process is repeated in the identical direction multiple times. The impurities become heavily concentrated at one end of the ingot, which is finally sawn off and discarded.
- **Applicable Elements: Silicon (Si), Germanium (Ge), Gallium (Ga), Indium (In), Boron (B)** — vital for transistor and semiconductor manufacturing!

2. Vapour Phase Refining: Two Essential Criteria

Vapour phase refining requires two strict chemical conditions:
1. The metal must react with an available reagent at low temperature to form a **readily volatile compound**.
2. The volatile compound must easily **decompose at a higher temperature to regenerate the pure metal**.

Process Name	Target Metal	Formation Reaction (Low Temp)	Decomposition Reaction (High Temp)
Mond's Process	**Nickel (Ni)**	**$Ni + 4CO \xrightarrow{(330-350\ K)} Ni(CO)_4(g)$**	**$Ni(CO)_4 \xrightarrow{(450-470\ K)} Ni(s) + 4CO(g)\uparrow$**
		Forms volatile, tetrahedral Nickel tetracarbonyl	Pure Ni deposits on heated pellets; CO recycled!
Van Arkel - de Boer Process	**Titanium (Ti), Zirconium (Zr), Hafnium (Hf), Boron**	**$Zr(s) + 2I_2(g) \xrightarrow{(870\ K)} ZrI_4(g)$** (Crude metal reacts in evacuated vessel)	**$ZrI_4(g) \xrightarrow{(W\ filament,\ 2075\ K)} Zr(s) + 2I_2(g)\uparrow$** Decomposes on white-hot filament; strips interstitial O_2/N_2!

Table 3.1 — Vapour phase refining pathways tested in JEE Advanced.

TRAP

TRAP: Van Arkel process is specifically designed to eliminate trace interstitial impurities of **Oxygen and Nitrogen** that cause severe embrittlement in space-grade Ti and Zr metals!

Electrolytic Refining (Electro-Refining) — Ultimate Purity for Copper, Silver & Gold

Electro-refining yields commercial copper (>99.95%) suitable for electrical wiring, where even 0.05% arsenic or iron reduces conductivity by >10%.

1. Electro-Refining Setup for Copper

Cell Component	Physical Material Used	Electrochemical Function / Mechanism
Anode (+)	Thick cast slabs of **crude Blister Copper** (≈98% pure)	Oxidizes copper and more electropositive base metals into solution: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
Cathode (–)	Very thin sheets of **pure Copper** coated with graphite	Selectively plates out pure copper metal from electrolyte: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
Electrolyte	Acidified aqueous $CuSO_4$ (≈15% $CuSO_4$ + 5% H_2SO_4)	Provides high Cu^{2+} ionic flux; H_2SO_4 prevents hydrolysis of Cu^{2+}
Operating Voltage	Low voltage (≈0.2 - 0.4 V) precisely controlled	Prevents more electropositive base metals from discharging at cathode

Table 3.2 — Electro-refining cell parameters.

2. Anode Mud (Anode Slime) — Chemistry & Economic Value

Impurities present in crude blister copper behave in two distinct ways based on their standard reduction potentials (E°):

- **More Electropositive Impurities (Fe, Zn, Ni):** Have more negative reduction potentials than copper ($E^\circ < +0.34$ V). They oxidize preferentially into the electrolyte as Fe^{2+}, Zn^{2+}, Ni^{2+} ions, but **DO NOT plate out at the cathode** because Cu^{2+} is much more readily reduced!
- **Less Electropositive / Noble Impurities (Ag, Au, Pt, Se, Te, Sb):** Have positive reduction potentials ($E^\circ > +0.34$ V). They are **NOT oxidized** at the low operating voltage and drop from the dissolving anode as an insoluble sludge called **ANODE MUD**!
- Economic Impact: The recovery of gold, silver, and platinum from anode mud completely covers the electrical cost of the entire refining plant!

KEY RESULT

Chromatographic Refining (NCERT §6.5): Based on differential adsorption of components of a mixture on an adsorbent (e.g. Al_2O_3 column). Used for separating micro-gram quantities of lanthanoids and actinoids.

Industrial Pyrometallurgy of Iron — The Blast Furnace (NCERT §6.6)

Iron is extracted from Haematite (Fe_2O_3) in a colossal vertical tower (blast furnace, ≈30 m high) lined with refractory fire-bricks, operating continuously for years under a counter-current flow of solids and hot gas.

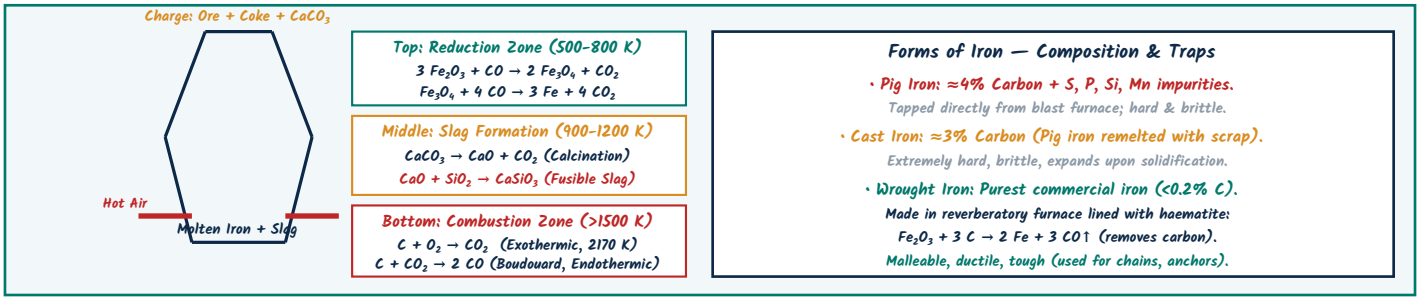


Fig. 3.2 — The Blast Furnace: Reduction zones, slag formation, combustion hearth, and forms of iron.

1. Thermal Zones & Stepwise Chemical Transformations

Zone & Temp Range	Primary Chemical Reactions	Thermodynamic Role / Nature
1. Zone of Reduction (Top of stack, 500 - 800 K)	$3\,Fe_2O_3 + CO \rightarrow 2\,Fe_3O_4 + CO_2$ $Fe_3O_4 + 4\,CO \rightarrow 3\,Fe + 4\,CO_2$ $Fe_2O_3 + CO \rightarrow 2\,FeO + CO_2$	Exothermic gas-solid reduction; CO is the primary reducing agent (Ellingham: CO → CO_2 lies lower)
2. Zone of Slag Formation (Belly / Middle, 900 - 1200 K)	$CaCO_3 \xrightarrow{(\Delta)} CaO + CO_2\uparrow$ (Calcination) $CaO + SiO_2 \rightarrow CaSiO_3$ (Slag formation) $FeO + CO \rightarrow Fe + CO_2$	Flux CaO captures quartz gangue SiO_2; Fusible molten $CaSiO_3$ floats above molten iron
3. Zone of Fusion & Carburization (Bosh, 1200 - 1500 K)	Spongy iron absorbs carbon (lowering its m.p. from 1811 K to ≈1473 K) and melts; $FeO + C \rightarrow Fe + CO$ (Direct carbon reduction)	Endothermic direct reduction by coke; Carbon dissolution creates Pig Iron (≈4% C)
4. Zone of Combustion (Hearth / Tuyeres, 1500 - 2170 K)	$C + O_2 \rightarrow CO_2$ ($\Delta H = -393.5$ kJ, T ≈ 2170 K) $C + CO_2 \rightarrow 2\,CO$ ($\Delta H = +172.5$ kJ, Boudouard)	Blasts of preheated air burn coke to CO_2, which immediately reduces to CO reducing gas

Table 3.3 — Comprehensive thermal and chemical zones of the blast furnace.

TRAP

TRAP: Does haematite get reduced directly to Fe in a single step? NO! Reduction proceeds strictly sequentially: $Fe_2O_3 \rightarrow Fe_3O_4 \rightarrow FeO \rightarrow Fe$!

Pyrometallurgy of Copper from Chalcopyrite — Matte Smelting & Auto-Reduction (NCERT §6.6)

Copper pyrites ($CuFeS_2$) contains equal molar amounts of copper and iron. The central metallurgical challenge is the **preferential removal of iron as slag** prior to reducing copper sulphide.

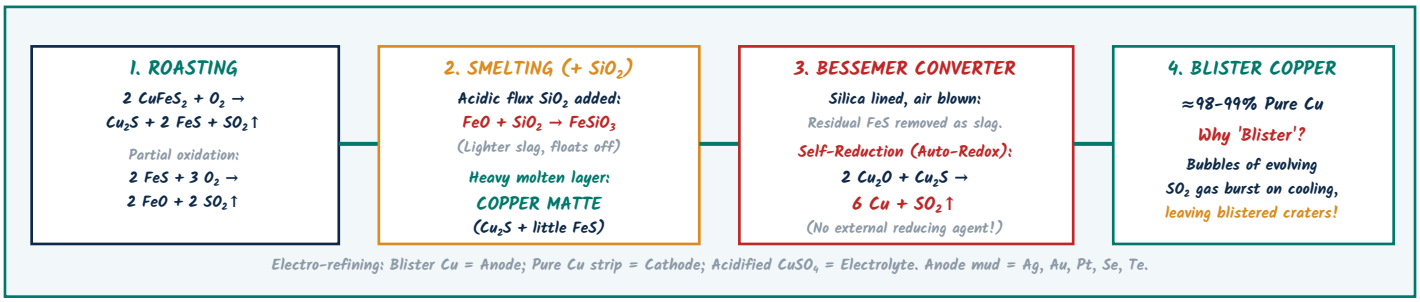


Fig. 3.3 — Copper pyrometallurgy: Roasting, smelting in reverberatory furnace, Bessemerization, and blister formation.

1. Stepwise Metallurgy of Copper Pyrites

- **Step 1: Crushing & Froth Floatation:** Concentrated using pine oil and potassium ethyl xanthate collector.
- **Step 2: Roasting in Reverberatory Furnace:** Heated in excess air to remove moisture, arsenic, and antimony as volatile oxides:
  $2\ CuFeS_2 + O_2 \rightarrow Cu_2S + 2\ FeS + SO_2 \uparrow\ 2\ FeS + 3\ O_2 \rightarrow 2\ FeO + 2\ SO_2 \uparrow$
- **Step 3: Smelting in Reverberatory Furnace (+ Silica Flux):** Roasted ore is mixed with **coke and acidic flux SiO_2**. Because Iron has a higher affinity for oxygen than copper, FeO preferentially reacts with SiO_2: **$FeO + SiO_2 \rightarrow FeSiO_3(l)$** (Fusible slag floats on top!)
  The heavier bottom molten layer is **COPPER MATTE** (primarily Cu_2S + trace FeS, ≈45-50% Cu).
- **Step 4: Bessemerization (Pear-shaped, silica-lined converter):** Hot air blast oxidizes residual FeS to FeO, captured by the silica lining as $FeSiO_3$ slag. Then, **auto-reduction** converts Cu_2S to molten copper:
  $2\ Cu_2S + 3\ O_2 \rightarrow 2\ Cu_2O + 2\ SO_2 \uparrow$ **$2\ Cu_2O + Cu_2S \rightarrow 6\ Cu(l) + SO_2 \uparrow$**
- **Step 5: Solidification:** Cooling molten copper releases dissolved SO_2 gas, creating blisters on the surface (**Blister Copper**, ≈98% pure). Refined by electro-refining.

TRAP
EXAMINER TRAP: Copper Matte composition is a universal favourite in JEE! Matte is a mixture of **Cu_2S and FeS** (NEVER CuS or Fe_2S_3!).

Hall-Héroult, Hydrometallurgy & Master Synthesis Matrix (NCERT §6.6 & §6.7)

A high-density master reference sheet consolidating aluminium electrometallurgy, zinc pyrometallurgy, gold/silver hydrometallurgy, and industrial metallurgical constants.

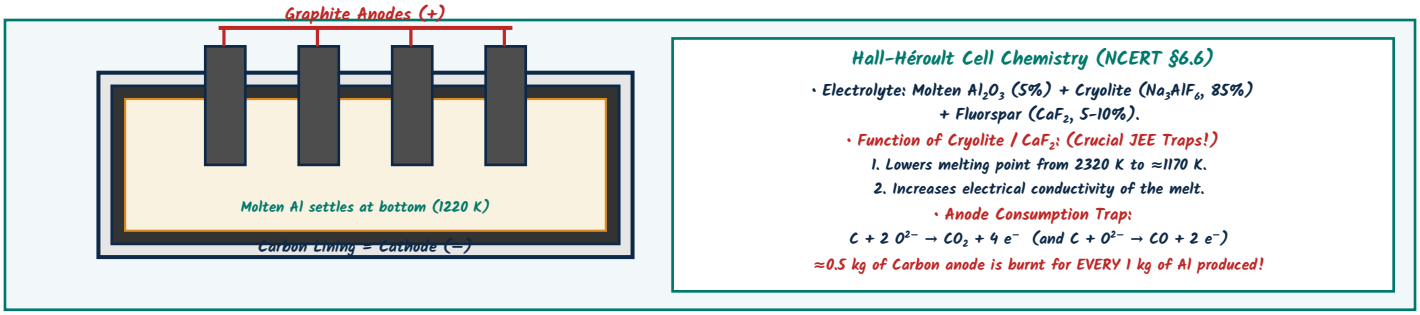


Fig. 3.4 — Hall-Héroult Electrolytic Cell: Carbon cathode lining, graphite anodes, and cryolite electrolyte.

1. Hall-Héroult Process Constants & Traps (NCERT §6.6)

- **Electrolyte Formulation:** Purified Al_2O_3 (≈5%) dissolved in molten **Cryolite (Na_3AlF_6, ≈85%)** and **Fluorspar (CaF_2, ≈10%)** at 1170-1220 K.
- **Why add Cryolite & CaF_2?** Pure alumina melts at an impractically high 2320 K and is an electrical insulator. Cryolite acts as a solvent, dramatically lowers the melting point to ≈1170 K, and provides mobile ions for high electrical conductivity!
- **Carbon Anode Consumption Trap:** At the anode, oxygen anions discharge to form CO and CO_2: $C + 2\,O^{2-} \rightarrow CO_2 + 4\,e^-$. For **EVERY 1 kg of aluminium produced, ≈0.5 kg of graphite anode is burned away**! Anodes must be continuously replaced.

2. Master Synthesis Matrix: Complete Extraction Pathways for JEE Advanced

Metal	Principal Ore	Beneficiation Method	Conversion to Oxide	Primary Reduction Process	Final Refining Method
Al	Bauxite ($Al_2O_3·2H_2O$)	Bayer's / Hall's Leaching	Calcination (1470 K)	**Hall-Héroult Electrolysis** (fused cryolite)	Hoope's Three-Layer Electrolytic Process
Fe	Haematite (Fe_2O_3)	Gravity wash + Magnetic	Calcination in kiln	**Carbon / CO Smelting** in Blast Furnace	Puddling furnace to wrought iron; Bessemer/BOS to steel
Cu	Chalcopyrite ($CuFeS_2$)	Froth Floatation (pine oil)	Partial Roasting	**Smelting + Self-Reduction** in Bessemer converter	**Electro-refining** (anode mud yields Au, Ag)
Zn	Sphalerite (ZnS)	Froth Floatation	Roasting to ZnO	**Carbon Reduction** in vertical retort (1673 K)	Fractional Distillation (removes Pb, Cd)
Pb	Galena (PbS)	Froth Floatation	Partial Roasting	**Self-Reduction** (reverberatory hearth)	Parke's / Pattinson's process (Ag removal) + Bett's process
Sn	Cassiterite (SnO_2)	Magnetic (removes wolframite)	Roasting (removes S, As)	**Carbon Smelting** in reverberatory furnace	Liquation followed by Poling
Au / Ag	Native Au / Argentite (Ag_2S)	Crushing & Pulverization	None (Direct leaching)	**Cyanide Hydrometallurgy + Zn displacement**	Cupellation (removes Pb) + Electrolytic refining

Table 3.4 — Master Extractive Metallurgy matrix across all metals for JEE Advanced.